A–B. Set Language, Constructions, and Regions

Elements, subsets, and equality

$x \in A$ means x is an element of A; $A \subseteq B$ means every element of A belongs to B. A proper subset $A \subset B$ also requires $A \neq B$. Sets are equal exactly when $A \subseteq B$ and $B \subseteq A$. Order and repetition do not matter inside a set.

Roster and set-builder notation

Roster notation lists elements; set-builder notation states a domain and a property. For example,

$$\{x \in \mathbb{Z} : -2 \leq x < 3\} = \{-2, -1, 0, 1, 2\}.$$

Read the colon as "such that." Always enforce the declared domain.

Power sets and Cartesian products

$\mathcal{P}(A)$ is the set of all subsets of A. If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. The empty set and A itself are included.

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Ordered pairs preserve order; generally $(a, b) \neq (b, a)$.

A worked construction

Let $S = \{x \in \mathbb{Z} : |x| \leq 2\}$ and $T = \{-1, 1\}$. Then

$$S = \{-2, -1, 0, 1, 2\}, \quad T \subseteq S, \quad T \notin S,$$

and

$$\mathcal{P}(T) = \{\varnothing, \{-1\}, \{1\}, \{-1, 1\}\}.$$

Step 1: solve $|x| \leq 2$ in the declared integer domain. **Step 2:** both elements of T occur in S, so $T \subseteq S$. **Step 3:** the elements of S are integers, so the set T is not itself an element: $T \notin S$. **Step 4:** list subsets of T by size (zero, one, then two elements); the count $4 = 2^2$ checks the power set.

Set operations

Relative to universe $\mathfrak{U}$,

$$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\}, \\ A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\ A \setminus B &= \{x : x \in A \text{ and } x \notin B\}, \\ \overline{A} &= \mathfrak{U} \setminus A. \end{aligned}$$

Membership formulas and set formulas describe the same regions.

Translate logic into a Venn region

Replace $x \in A$ by A, $\vee$ by union, $\wedge$ by intersection, and $\neg$ by complement. First eliminate conditionals:

$$P \to Q \equiv \neg P \vee Q, \qquad P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q).$$

Then shade exactly the resulting region.

B worked translations

$$\begin{aligned} (x \in A \setminus B) \vee (x \in C) &\longleftrightarrow (A \setminus B) \cup C, \\ (x \in A \cup B) \to (x \in C) &\longleftrightarrow \overline{A \cup B} \cup C, \\ (x \in A) \leftrightarrow (x \in B \setminus C) &\longleftrightarrow [A \cap (B \setminus C)] \\ &\quad \cup [\overline{A} \cap \overline{(B \setminus C)}]. \end{aligned}$$

Procedure: (1) eliminate $\to$ or $\leftrightarrow$; (2) replace $\neg, \wedge, \vee$ by complement, intersection, and union; (3) simplify. For the biconditional, the first term is the region where both sides are true; the second is where both are false. Shade both.

Common errors

$\{x\} \in \mathcal{P}(A)$ is equivalent to $x \in A$, but $\{x\} \in A$ is a different claim. Include $\varnothing$ in every power set. Do not treat a Cartesian product as unordered.

A–B readiness

Can you move between roster and set-builder notation; distinguish $\in$ from $\subseteq$; build $\mathcal{P}(A)$ and $A \times B$; and convert a compound membership formula into a set expression and Venn region?

C. Relations as Subsets of Cartesian Products

Relations and plots

A relation from X to Y is any subset $R \subseteq X \times Y$. A plotted point at horizontal coordinate x and vertical coordinate y represents $(x, y) \in R$. The **domain of the relation** is the set of first coordinates that occur; its **range** is the set of second coordinates.

When a relation is a function

A relation $R \subseteq X \times Y$ is a function $X \to Y$ exactly when

$$\forall x \in X \; \exists! y \in Y \; ((x, y) \in R).$$

Every input must have exactly one output. An input with no point or with two points proves the relation is not a function.

Injective and surjective

For a function $f : X \to Y$:

$$\begin{aligned} \text{injective}: &\quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2, \\ \text{surjective}: &\quad \forall y \in Y \; \exists x \in X \; (f(x) = y), \\ \text{bijective}: &\quad \text{both injective and surjective.} \end{aligned}$$

On a finite plot, injective means no two inputs share an output; surjective means every codomain value is hit.

C worked plot-to-pairs example

Let $X = Y = \{1, 2, 3\}$ and suppose the plot has points

$$R = \{(1, 2), (2, 3), (3, 2)\}.$$

Function check: columns $x = 1, 2, 3$ each contain exactly one point, so R is a function $X \to Y$. **Injective check:** inputs 1 and 3 both map to 2, so it is not injective. **Surjective check:** codomain value $1 \in Y$ is never hit, so it is not surjective.

Read quantified plot statements

$$\begin{aligned} \forall x \exists y \, R(x, y): &\quad \text{every column has a point,} \\ \forall x \exists! y \, R(x, y): &\quad \text{every column has exactly one,} \\ \forall y \exists x \, R(x, y): &\quad \text{every row has a point,} \\ \forall y \exists! x \, R(x, y): &\quad \text{every row has exactly one,} \\ \exists x \forall y \, R(x, y): &\quad \text{some column is full,} \\ \exists y \forall x \, R(x, y): &\quad \text{some row is full.} \end{aligned}$$

The first variable is the horizontal coordinate in (x, y).

C quantified evaluation

For $R = \{(1, 2), (2, 3), (3, 2)\}$:

- $\forall x \exists! y \, R(x, y)$: scan every column; each has one point, so true.
- $\forall y \exists x \, R(x, y)$: row $y = 1$ is empty, a counterexample, so false.
- $\exists y \forall x \, R(x, y)$: test each row; none contains all three inputs, so false.

For a false universal, name the missing row or column; for a true existential, name the row or column that is the witness.

C nonfunction diagnosis

If $S = \{(1, 1), (1, 3), (2, 2), (3, 3)\}$, then input 1 has two outputs. Thus S is not a function $X \to Y$, even though every input has at least one output. Classification as injective or surjective is requested only after the relation passes the function test.

Common errors

Do not confuse "exactly one output for each input" with "exactly one input for each output." A function may be many-to-one. Surjectivity is always relative to the declared codomain, not merely the values visible in a formula.

C readiness

Can you list a plotted relation as ordered pairs, evaluate each quantifier pattern, identify a missing or duplicated output, and classify a valid function as injective, surjective, both, or neither?

D. Functions, Images, and Preimages

Domain, codomain, range

In $f : X \to Y$, X is the domain and Y the codomain. The range is $f(X) = \{f(x) : x \in X\}$ and may be smaller than Y. The same formula can have different injectivity or surjectivity behavior when its declared domain or codomain changes.

Images and preimages of sets

For $U \subseteq X$ and $P \subseteq Y$,

$$f(U) = \{f(x) : x \in U\}, \qquad f^{-1}(P) = \{x \in X : f(x) \in P\}.$$

$f^{-1}(P)$ is preimage notation and does not require f to have an inverse function.

D finite-function example

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, and $f(1) = a, f(2) = b, f(3) = b, f(4) = c$. Then

$$f(\{1, 3, 4\}) = \{a, b, c\}, \quad f^{-1}(\{a, b\}) = \{1, 2, 3\}.$$

Image: evaluate inputs $1, 3, 4$ to get a, b, c. **Preimage:** collect all inputs whose outputs lie in $\{a, b\}$, namely $1, 2, 3$. **Classification:** every element of B is hit, so f is surjective; $f(2) = f(3) = b$ with $2 \neq 3$, so it is not injective.

Proving or disproving injectivity

To prove injective, start with arbitrary x_1, x_2 and assume $f(x_1) = f(x_2)$; derive $x_1 = x_2$. To disprove it, give distinct inputs with the same output.

Proving or disproving surjectivity

To prove surjective, let arbitrary y lie in the codomain and solve $f(x) = y$ for an allowed x in the domain. To disprove it, identify one codomain element with no preimage.

D integer formula examples

For $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = n + 5$:

$$g(\{-1, 0, 2\}) = \{4, 5, 7\}, \quad g^{-1}(\{4, 7\}) = \{-1, 2\}.$$

Injective: assume $g(n_1) = g(n_2)$; then $n_1 + 5 = n_2 + 5$, hence $n_1 = n_2$. **Surjective:** let arbitrary $y \in \mathbb{Z}$; choosing $n = y - 5 \in \mathbb{Z}$ gives $g(n) = (y - 5) + 5 = y$. Thus g is bijective.

Domain and codomain matter

For $h : \mathbb{Z} \to \mathbb{Z}$, $h(n) = n^2$, the inputs -1 and 1 collide, so h is not injective. No integer squares to -1, so it is not surjective. For $q : \mathbb{N}_0 \to \{m^2 : m \in \mathbb{N}_0\}$, $q(n) = n^2$, the domain removes the $n, -n$ collisions. Every declared codomain value has the form $m^2 = q(m)$ for an allowed input. Thus q is injective and surjective. The formula did not change; the declared domain and codomain did.

Common errors

An image removes duplicate outputs. A preimage collects inputs, not output values. Never classify a function without reading both domain and codomain. Do not claim surjectivity by saying "the formula has outputs"; construct a preimage for an arbitrary codomain element.

D readiness

Can you compute images and preimages; distinguish range from codomain; prove or disprove injectivity and surjectivity with the correct witnesses; and explain how changing the domain or codomain changes the classification?

E. Prove or Refute Set and Function Claims

Element-chasing proof template

To prove $S = T$, prove both inclusions. For $S \subseteq T$:

1. Let $z \in S$ be arbitrary.
2. Unpack membership in S using definitions.
3. Apply the relevant set/function definitions.
4. Conclude $z \in T$.

Do not begin by assuming the desired equality.

True claim: image of a union

For $f : X \to Y$ and $U, V \subseteq X$, $f(U \cup V) = f(U) \cup f(V)$. **Forward inclusion:** if $y \in f(U \cup V)$, then $y = f(x)$ for some $x \in U \cup V$. Thus $x \in U$ or $x \in V$, so $y \in f(U)$ or $y \in f(V)$. This proves one inclusion. **Reverse inclusion:** if $y \in f(U) \cup f(V)$, then $y = f(x)$ for some $x \in U$ or some $x \in V$; in either case $x \in U \cup V$, so $y \in f(U \cup V)$. Both inclusions prove equality.

False claim: image of an intersection

In general, $f(U \cap V) \neq f(U) \cap f(V)$. Let $X = \{1, 2\}$, $Y = \{a\}$, $f(1) = f(2) = a$, $U = \{1\}$, and $V = \{2\}$. Then

$$f(U \cap V) = f(\varnothing) = \varnothing, \quad f(U) \cap f(V) = \{a\}.$$

Verification: the hypotheses hold, but the left side is empty and the right side is not, so the proposed universal equality is false. It fails because two different inputs may have the same output.

True claim: preimage preserves intersection

For $P, Q \subseteq Y$,

$$f^{-1}(P \cap Q) = f^{-1}(P) \cap f^{-1}(Q).$$

Indeed, for arbitrary $x \in X$,

$$\begin{aligned} x \in f^{-1}(P \cap Q) &\iff f(x) \in P \cap Q \\ &\iff f(x) \in P \text{ and } f(x) \in Q \\ &\iff x \in f^{-1}(P) \cap f^{-1}(Q). \end{aligned}$$

The chain proves both inclusions at once.

True claim: product distributes over union

For arbitrary (x, y),

$$\begin{aligned} (x, y) \in (A \cup B) \times C &\iff (x \in A \text{ or } x \in B) \text{ and } y \in C \\ &\iff [(x \in A \text{ and } y \in C) \text{ or} \\ &\qquad (x \in B \text{ and } y \in C)] \\ &\iff (x, y) \in (A \times C) \cup (B \times C). \end{aligned}$$

False cancellation claim

$A \cup B = A \cup C$ need not imply $B = C$. Take $A = \{1, 2\}$, $B = \{1\}$, and $C = \{2\}$. Then both unions equal $\{1, 2\}$, so the hypothesis is true, but $B \neq C$, so the conclusion is false. Union with A hides differences already contained in A. A valid counterexample must pass both of these checks.

Common errors

Examples do not prove universal claims. For a false equality, compute both sides of a concrete finite example. In an image proof, different witnesses may map to the same output; do not silently identify them.

Whole-Core readiness

Can you construct finite sets, translate Venn regions, interpret relations, compute and classify functions, and choose between an arbitrary-element proof and a concrete counterexample? A live 20-minute form selects one complete problem from each section A–E.